Two Consecutive Loops

Topic: TimeComplexity File: p3.pdf

PROBLEM

Find the complexity when one loop runs n times and a second, separate loop also runs n times.

PSEUDOCODE

```
for i = 0 to n - 1: process(A[i])
for j = 0 to n - 1: process(B[j])
```

ANALYSIS

The first loop performs n operations and the second performs n. The total is n + n = 2n.

EXAMPLE

For n = 100, there are about 200 constant-time loop-body executions, not 10000.

COMPLEXITY

Time: O(n), because constant factors are ignored by asymptotic notation. Extra space: O(1).

TAKEAWAY

Sequential blocks add their costs; nested blocks often multiply their iteration counts.